VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS302

Database Management Systems & Distributed Transactions - Semester Exam Q

Syllabus Focus: CS302 High-Yield Semester Exam Question Bank & Model Solutions

Summary: Comprehensive question bank with previous semester university examination questions, m

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Relational Formalisms & Schema Design

" Syllabus Scope: Relational algebra operators, tuple relational calculus, ER-to-relational schema mappin

Unit 2: Structured Query Language SQL

" Syllabus Scope: Complex joins, correlated subqueries, aggregate grouping, window functions, views, trig

CS302: Database Management Systems & Distributed Transactions - Advanced Topics &

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Functional Dependencies & Normalization

" Armstrong axioms, canonical cover, attribute closure, 1NF, 2NF, 3NF, BCNF, lossless join, dependency preserv

Unit 4: Transaction ACID & Concurrency Control

" Serializability, conflict graph testing, Two-Phase Locking 2PL, Strict 2PL, timestamp ordering, deadlock det

Unit 5: Storage Engines, Indexing & Query Tuning

" B+ Tree index insertion and deletion, hash indexes, query execution trees, cost-based optimizer estimates

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Given relation RA,B,C,D,E and FDs {A->B, BC->D, D->E, E->A}, compute all candidate keys and decompose
2. Construct a precedence graph for a concurrent transaction schedule of 4 transactions to determine conf
3. Trace the node splitting operations of a B+ Tree of order 3 when inserting keys [10, 20, 30, 40, 50, 6

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Raghu Ramakrishnan, Johannes Gehrke - Database Management Systems McGraw-Hill 3rd Ed
* Abraham Silberschatz, Henry F. Korth - Database System Concepts McGraw-Hill 7th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students